

Anmerkung - Exact Executable Condition Table

Current rule inventory extracted from the executable processor logic in **assets/anmerkung.js**. This report describes the conditions that are actually evaluated, including precedence, thresholds, code-book mappings, terminal returns, and disabled branches.

Repository	nicoyogi/vb-code-vault	Source file	assets/anmerkung.js
Source blob SHA	a40e79f662b910198f6bf6e3841329e728b2d7fe	Checked	2026-09-14
Processors	processDachser / processKN / processDHL / processWackler	Core numeric guard	hasErr(v,t) = abs(v) > t

Carrier thresholds

Forwarder	Constant	Default	Exact numeric condition
Dachser	T_DACHSER	0.08	hasErr(v,t) => abs(v) > 0.08
K+N	T_KN	0.09	hasErr(v,t) => abs(v) > 0.09
DHL	T_DHL	0.04	hasErr(v,t) => abs(v) > 0.04
Wackler	T_WACKLER	0.09	hasErr(v,t) => abs(v) > 0.09

Important current-state note: **KONTIERUNG_ENABLED=false**. The code contains Kontierung? branches, but they are dormant in the current executable configuration.

1. Dachser - processor rules

FW	Rule	Input / field	Exact condition -> Anmerkung output
Dachser	Gate	Stat_Freigabe	Condition: cellNum(stat) != 10 Output: Return null; no Anmerkung processing
Dachser	1	C502_DL	Condition: raw value exists, raw != "-" and cellNum != 0 Output: Einlagern
Dachser	2	C503_DL	Condition: raw value exists, raw != "-" and cellNum != 0 Output: Auslagern
Dachser	3	LG Differenz	Condition: abs(value) > T_DACHSER Output: Lagergeld
Dachser	4	AV Differenz	Condition: abs(value) > T_DACHSER Output: Gebühr für vergeblichen Abholversuch
Dachser	5	ZZ Differenz	Condition: abs(value) > T_DACHSER Output: 2. Zustellung
Dachser	6	SNK	Condition: See dedicated SNK cascade below Output: SNK-derived phrase
Dachser	7	SAM Differenz	Condition: abs(value) > T_DACHSER Output: Samstagzustellung
Dachser	Early exit	TARIF	Condition: TARIF blank + FR significant Output: Special blank-tariff branches below; then stop only for tariff-zero marker
Dachser	8	DGR Differenz	Condition: abs(value) > T_DACHSER Output: Gefahrgut-Zuschlag
Dachser	9	EXP Differenz	Condition: abs(value) > T_DACHSER Output: If EXP_DL == 95: Terminzuschlag; else Produktzuschlag
Dachser	10	MT Differenz	Condition: abs(value) > T_DACHSER Output: Mautdifferenz
Dachser	11	Empf.-Ort/PLZ	Condition: Ort == LONDON and Empf.-PLZ starts with A-Z Output: Zone korrekt? plus Einfuhrzollabfertigung unless EXP diff is significant
Dachser	12	SBfU Differenz	Condition: abs(value) > T_DACHSER Output: SBfU-Bescheinigung f. Umsatzsteuerzwecke
Dachser	13	FR Differenz	Condition: abs(FR) > T OR (abs(FR) >= 0.05 AND no result yet) Output: Enter FR classification cascade below
Dachser	14	SNK_DL / SNK Differenz	Condition: SNK_DL == 14 AND abs(SNK_DIFF) > T Output: Abholterminvereinbarung
Dachser	15	38L Differenz	Condition: abs(value) > T_DACHSER Output: Maut von/bis NL
Dachser	16	TZ Differenz	Condition: result empty AND no FR trigger AND abs(TZ) > T_DACHSER Output: Differenz treibstof

1A. Dachser - SNK code cascade

FW	Rule	Input / field	Exact condition -> Anmerkung output
Dachser	SNK-1	SNK_DL	Condition: effective code 190 Output: AUSFALLFRACHT
Dachser	SNK-2	SNK_DL	Condition: effective code 95 Output: AUSFALLFRACHT
Dachser	SNK-3	SNK_DL	Condition: effective code 130 Output: Standgeld
Dachser	SNK-4	SNK_DL=75	Condition: SERV_ART == K1AV Output: Speditionskosten gem. Text
Dachser	SNK-5	SNK_DL=75	Condition: SNK_TARIF > 0 AND abs(SNK_DIFF) > T Output: Differenz Hebebuehnen-Zuschlag
Dachser	SNK-6	SNK_DL=75	Condition: SNK_TARIF == 0/blank Output: Ausfallfracht/Schadensersatz
Dachser	SNK-7	SNK_DL=11	Condition: abs(SNK_DIFF) <= T Output: No SNK phrase
Dachser	SNK-8	SNK_DL=11	Condition: abs(SNK_DIFF) > T Output: Differenz Telefonische Zustellankündigung - Laderaumzuschlag
Dachser	SNK-9	SNK_DL=14	Condition: abs(SNK_DIFF) <= T OR TARIF is zero marker Output: No SNK phrase

FW	Rule	Input / field	Exact condition -> Anmerkung output
Dachser	SNK-10	SNK_DL=14	Condition: abs(SNK_DIFF) > T AND TARIF not zero marker AND SERV_ART=K1AV Output: Differenz Laderaumkostenentwicklung
Dachser	SNK-11	SNK_DL=14	Condition: abs(SNK_DIFF) > T AND TARIF not zero marker AND SERV_ART!=K1AV Output: Differenz Automatische Zustellterminvereinbarung - Laderaumzuschlag
Dachser	SNK-12	SNK_DL=5	Condition: abs(SNK_DIFF) <= T Output: No SNK phrase
Dachser	SNK-13	SNK_DL=5	Condition: abs(SNK_DIFF) > T AND SERV_ART=K1AV AND SNK_TARIF=0 Output: Admin Zeitfensterbuchung Handel
Dachser	SNK-14	SNK_DL=5	Condition: abs(SNK_DIFF) > T AND SERV_ART=K1AV AND SNK_TARIF!=0 Output: Differenz Admin Zeitfensterbuchung Handel - Laderaumzuschlag
Dachser	SNK-15	SNK_DL=5	Condition: abs(SNK_DIFF) > T AND SERV_ART!=K1AV AND TARIF not zero Output: Differenz Automatische Zustellterminvereinbarung - Laderaumzuschlag
Dachser	SNK-16	SNK_DL=9	Condition: abs(SNK_DIFF) <= T OR TARIF zero Output: No SNK phrase
Dachser	SNK-17	SNK_DL=9	Condition: abs(SNK_DIFF) > T AND TARIF not zero Output: Differenz Telefonische Zustellterminvereinbarung - Laderaumzuschlag
Dachser	SNK-18	default	Condition: abs(SNK_DIFF) <= T Output: No SNK phrase
Dachser	SNK-19	default	Condition: abs(SNK_DIFF) > T AND SERV_ART=K1AV Output: Differenz Laderaumkostenentwicklung
Dachser	SNK-20	default	Condition: abs(SNK_DIFF) > T AND SNK_DL non-integer AND SNK_DIFF non-integer Output: Differenz Laderaumkostenentwicklung
Dachser	SNK-21	default	Condition: abs(SNK_DIFF) > T otherwise Output: Differenz Automatische Zustellterminvereinbarung - Laderaumzuschlag
Dachser	SNK-derivate	non-integer SNK_DL	Condition: rounded SNK_DIFF in {5,9,11,14} and within 0.05 Output: Use rounded code as effective SNK_DL

1B. Dachser - FR classification / precedence

FW	Rule	Input / field	Exact condition -> Anmerkung output
Dachser	FR-1	FR gate	Condition: Anz_Sdg > 1 Output: hätte gebündelt werden können?
Dachser	FR-2	FR gate	Condition: Anz_Sdg <= 1 AND REFERENZ3 == ZW Output: Differenz aufgrund abweichender Zwischenempfänger [KI_ZW_PLZ/ORT or fallback Empf.-PLZ/Ort]
Dachser	FR-3	FR gate	Condition: SERV_ART == K1AS Output: Sonderfahrt
Dachser	FR-4	FR gate	Condition: SERV_ART == K1AU AND abs(FR-90) <= 0.1 Output: Sonderfahrt duplicate-90 phrase
Dachser	FR-5	known weights	Condition: same weight OR same tariff tier; FR > 1 OR (FR >= 0.05 AND no prior result) Output: Fracht Differenz (ZZ=35 variant: "Fracht Differenz"; otherwise "Frachtdifferenz")
Dachser	FR-6	known weights	Condition: same weight/tier; FR < -1.0 AND no prior result Output: Differenz aufgrund abweichender Gewichte
Dachser	FR-7	known weights, cross-tier	Condition: 0 < FR < 1.0 Output: Fracht Differenz / Frachtdifferenz depending ZZ==35
Dachser	FR-8	known weights, cross-tier	Condition: otherwise Output: Differenz aufgrund abweichender Gewichte
Dachser	FR-9	weights missing	Condition: FR < 0 AND (abs(FR)<1.0 OR VKG_DL > 0) Output: Differenz Frachtzu/ abschlag
Dachser	FR-10	weights missing	Condition: otherwise Output: Differenz aufgrund von abweichendem Gewicht
Dachser	ZW prece dence	REFERENZ3=ZW	Condition: Anz_Sdg > 1 Output: Bundling phrase wins over Zwischenempfänger note
Dachser	Tarif-zero foreign	blank TARIF	Condition: FR significant AND Abg.-Land present and != DE Output: kein Tarif für <Abg.-Land>
Dachser	Tarif-zero Vorholung	blank TARIF	Condition: FR significant AND Abg.-Land not foreign AND no SACH/SERV and no other diff/surcharge Output: VORHOLUNG
Dachser	Tarif-zero Fremdnu mmer	blank TARIF	Condition: FR significant AND SACHKONTO and SERV_ART populated Output: Fremdnummer 5034xxx bereits berechnet in RE00123xxx, ok?
Dachser	Tarif-zero guards	blank TARIF	Condition: FR significant but TZ or MT etc. significant Output: Do not emit VORHOLUNG; continue if branch does not early-return

Dachser FR weight-tier reference

Tier ceiling table used by dachserGetTier(): 50, 100, 150, 200, 250, 300, 350, 400, 450, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700, 1800, 1900, 2000, 2200, 2400, 2600, 2800, 3000, 3500, 4000, 4500, 5000, 5500, 6000, 6500, 7000, 7500, 8000, 8500, 9000, 9500, 10000, 999999 kg. Same-tier means both weights map to the same ceiling breakpoint.

2. K+N - processor rules

FW	Rule	Input / field	Exact condition -> Anmerkung output
K+N	Gate	Stat_Freigabe	Condition: cellNum(stat) != 10 Output: Return null
K+N	FR-1	FR Differenz	Condition: abs(FR)>T AND ReferenzNr contains comma Output: Immediate return: hätte gebündelt werden müssen, ok?
K+N	FR-2	FR Differenz	Condition: abs(FR)>T AND TARIF blank or "-" Output: Pauschalfracht
K+N	FR-3	FR Differenz	Condition: abs(FR)>T AND recipient contains amazon AND no multi-ref AND VKG tiers differ Output: Differenz aufgrund abweichender Gewichte
K+N	FR-4	FR Differenz	Condition: abs(FR)>T AND recipient contains amazon AND no multi-ref AND weights absent/same tier Output: hätte nach Amazon Tarif abrechnen müssen
K+N	FR-5	FR Differenz	Condition: FR>0 AND SNK_DL == 5 AND same VKG/VKG_DL tariff tier Output: hätte nach Amazon Tarif abrechnen dürfen
K+N	FR-6	FR Differenz	Condition: FR>0 AND SNK_DL == 5 AND cross-tier Output: Differenz aufgrund abweichender Gewichte
K+N	FR-7	FR Differenz	Condition: abs(FR)>T otherwise Output: Differenz aufgrund abweichender Gewichte
K+N	EXP	EXP Differenz	Condition: abs(EXP)>T Output: FIXTERMIN
K+N	MT	MT Differenz	Condition: abs(MT)>T Output: Mautdifferenz
K+N	SNK-1	SNK Differenz	Condition: abs(SNK_DIFF-9)<0.01 Output: Avis, ok?
K+N	SNK-2	SNK Differenz	Condition: abs(SNK_DIFF+9)<0.01 Output: Differenz avis
K+N	SNK-3	SNK Differenz	Condition: abs(abs(SNK_DIFF)-25)<0.01 Output: Portalavisierung, ok?
K+N	SNK-4	SNK-DL mapping	Condition: abs(SNK_DIFF)>T AND SNK_DL in {5,25} Output: Portalavisierung, ok?
K+N	SNK-5	SNK-DL mapping	Condition: abs(SNK_DIFF)>T AND SNK_DL=9 Output: Avis, ok?
K+N	SNK-6	SNK-DL mapping	Condition: abs(SNK_DIFF)>T AND SNK_DL=12 Output: B2C-zuschlag, ok?
K+N	SNK-7	SNK-DL mapping	Condition: abs(SNK_DIFF)>T AND SNK_DL=18 AND no Avis phrase yet Output: Avis, ok?
K+N	SNK-8	SNK-DL mapping	Condition: abs(SNK_DIFF)>T AND SNK_DL=34 Output: Portalavisierung, ok? + Avis, ok?
K+N	SNK-9	SNK fallback	Condition: abs(SNK_DIFF)>T otherwise Output: SNK Differenz
K+N	TZ	TZ Differenz	Condition: abs(TZ)>T AND FR not significant AND MT not significant Output: Differenz treibstoff
K+N	Kontierung (disabled)	KOST/SACH	Condition: KONTIERUNG_ENABLED && either blank/- Output: Would append Kontierung?; currently KONTIERUNG_ENABLED=false

3. DHL Express - processor rules

FW	Rule	Input / field	Exact condition -> Anmerkung output
DHL	Gate	Stat_Freigabe	Condition: cellNum(stat) != 10 Output: Return null
DHL	1	TARIF	Condition: raw non-empty AND numeric value == 0 AND raw contains "0" or is "-" Output: Fremdnummer xxx bereits berechnet in RExxx, ok?.
DHL	2	KOST/SACH	Condition: KONTIERUNG_ENABLED && SACH blank Output: kontierung?
DHL	3	KOST/SACH	Condition: KONTIERUNG_ENABLED && KOST blank Output: kontierung?
DHL	4	FR Differenz	Condition: abs(value)>T Output: Differenz aufgrund von abweichendem Gewicht/Volumen (sets block=true)
DHL	5	PAL Differenz	Condition: abs(value)>T Output: nicht stapelbar ok? (sets block=true)
DHL	6	OW Differenz	Condition: abs(value)>T Output: overweight ok? (sets block=true)
DHL	7	YO Differenz	Condition: YO>0 AND YO % 15 == 0 Output: Non conveyable piece-weight ok? (sets block=true)
DHL	8	YO Differenz	Condition: abs(YO)>T otherwise Output: non conveyable piece ok? (sets block=true)
DHL	9	YL Differenz	Condition: abs(value)>T Output: Non-conveyable piece irregular ok? (sets block=true)
DHL	10	ND Differenz	Condition: abs(value)>T Output: Neutral delivery ok? (sets block=true)
DHL	11	SF Differenz	Condition: abs(value)>T Output: Direct signature ok? (sets block=true)
DHL	12	SNK Differenz	Condition: SNK == 25 Output: Limited quantities ok? (sets block=true)
DHL	13	SNK Differenz	Condition: SNK == 30 Output: Elevated Risk, ok? (sets block=true)
DHL	14	SNK Differenz	Condition: SNK == 60 Output: Eelevated risk ok? // Restricted destination ok? (sets block=true)
DHL	15	SNK Differenz	Condition: abs(SNK)>T otherwise Output: SNK Differenz (sets block=true)
DHL	16	AC Differenz	Condition: only when block=false; AC == 11 Output: Addres Correction, ok?
DHL	17	AC Differenz	Condition: only when block=false; abs(AC)>T otherwise Output: Address Correction ok?
DHL	18	MT Differenz	Condition: only when block=false; abs(MT)>T Output: Mautdifferenz
DHL	19	NX Differenz	Condition: only when block=false; abs(NX)>T Output: demand surcharge ok?
DHL	20	OS Differenz	Condition: only when block=false; abs(OS)>T Output: Oversize piece ok?
DHL	21	TZ Differenz	Condition: result empty AND abs(TZ)>T Output: Differenz treibstof

4. Wackler - processor rules

FW	Rule	Input / field	Exact condition -> Anmerkung output
Wackler	Gate	Stat_Freigabe	Condition: stat != 10 Output: Return null (Kontierung? path is dormant because flag=false)
Wackler	1	TARIF	Condition: "-" OR numeric 0 with non-empty raw OR blank TARIF with significant FR/MT/TZ Output: Fremdnummer xxx bereits berechnet in RExxx, ok?; suppress rules 3-12
Wackler	1b	TARIF+SNK	Condition: TARIF blank AND SNK significant AND no FR/MT AND TZ <2 AND no AVIS code AND SNK not recognized Output: Lagergeld, ok?; terminal return
Wackler	2	Pauschal	Condition: SNK significant, no FR/MT, TZ <2, TARIF>0, SNK >=TARIF, SNK unrecognized, not weighted-80, no AVIS code Output: Pauschalfracht, ok?
Wackler	3	AVIS	Condition: abs(AVIS) within 0.01 of 7.5, 8.5, 6.5 or 8.7 Output: Avis, ok?
Wackler	3a	AVIS	Condition: AVIS is negative -8.7 (within 0.01) Output: hätte Avisgebühr telefonisch abrechnen dürfen
Wackler	4	SNK code	Condition: abs(SNK) within tolerance of a recognized code (table below) Output: Code-specific label
Wackler	5A	weights	Condition: both VKG and VKG_DL >0 AND abs(FR)>T AND colli>=7 Output: High-colli decision tree below
Wackler	5B	weights	Condition: both VKG and VKG_DL >0 AND abs(FR)>T AND colli<7 AND same tier Output: Wackler rechnet Frachtrate für <reportTier>kg ab
Wackler	5C	weights	Condition: both weights >0 AND abs(FR)>T AND cross-tier AND multi-ref AND VKG_DL<=20%*VKG Output: hätte gebündelt werden müssen; terminal return
Wackler	5D	weights	Condition: cross-tier AND multi-ref AND FR>0 AND weight gap<=20% of max Output: hätte gebündelt werden müssen
Wackler	5E	weights	Condition: cross-tier otherwise Output: Differenz aufgrund abweichender Gewichte
Wackler	5F	no weights	Condition: no weight-triggered branch AND multi-ref AND FR significant Output: hätte gebündelt werden müssen
Wackler	6	AVIS	Condition: abs(AVIS-1)<0.01 Output: Differenz avis, ok?
Wackler	7	Empf.-PLZ/Ort	Condition: PLZ == 88499 AND Ort == RIEDLINGEN Output: Return, ok?
Wackler	8	FR fallback	Condition: abs(FR)>T AND no weight/bundle trigger AND no "gebündelt" and no "return" in result Output: Frachtdifferenz
Wackler	9	MT	Condition: abs(MT)>T Output: Mautdifferenz (always additive)
Wackler	10	SNK fallback	Condition: SNK unrecognized AND abs(SNK)>=5 AND not bundled Output: If SNK<0 and abs(SNK -150)<=2: Hebebuehne; else SNK Differenz
Wackler	11	TZ	Condition: abs(TZ)>=2 AND not wacklerRechnetFired AND not FR credit Output: DifferenzEnergiezuschlag; if Hebebuehne fired: Dieselzuschlag
Wackler	12	NL-FIX corollary	Condition: SNK code == NL-FIX AND FR significant AND KOST/SACH both blank or X Output: Prefix Zone korrekt? and remove Frachtdifferenz
Wackler	13 (disabled)	KOST/SACH	Condition: KONTIERUNG_ENABLED && both blank/X Output: Would append Kontierung?; currently disabled
Wackler	14	TZ fallback	Condition: result empty AND abs(TZ)>T Output: DifferenzEnergiezuschlag

4A. Wackler - SNK code book

FW	Rule	Input / field	Exact condition -> Anmerkung output
Wackler	SNK	signature	Condition: 38 ±0.5 Output: NL-FIX
Wackler	SNK	signature	Condition: 11.5 ±0.1 Output: hätte B2C-Line abrechnen dürfen
Wackler	SNK	signature	Condition: 22 ±0.5 Output: 2. Zustellung ok?
Wackler	SNK	signature	Condition: 25 ±0.5 Output: Terminzustellung
Wackler	SNK	signature	Condition: 170 ±0.5 Output: Terminzustellung
Wackler	SNK	signature	Condition: 180 ±0.5 Output: Terminzustellung, ok?
Wackler	SNK	signature	Condition: 43 ±0.5 Output: 2.Zustellung ok?
Wackler	SNK	signature	Condition: 289 ±0.5 Output: Umverfügung
Wackler	SNK	signature	Condition: 49 ±0.5 Output: NL-12, ok?
Wackler	SNK	signature	Condition: 113 ±0.5 Output: NL-SPEZ, ok?
Wackler	SNK/weight-gated	signature	Condition: 80 ±0.5 AND VKG>0 AND VKG_DL>0 Output: Terminzustellung

4B. Wackler - weight / bundling decision tree

FW	Rule	Input / field	Exact condition -> Anmerkung output
Wackler	High-colli	signature	Condition: colli>=7; bundle; real-weight tier gap >1% and <=5% Output: hätte gebündelt werden müssen
Wackler	High-colli	signature	Condition: colli>=7; real VKG/VKG_DL tiers differ Output: Differenz aufgrund abweichender Gewichte
Wackler	High-colli	signature	Condition: colli>=7; same real tier; refCount>=6 Output: hätte gebündelt werden müssen
Wackler	High-colli	signature	Condition: colli>=7; same real tier; palletVol=colli*285 materially exceeds VKG (>=5% or >10000) AND chargeable tier differs from DL tier Output: Differenz aufgrund abweichender Gewichte
Wackler	High-colli	signature	Condition: colli>=7; same real tier; refCount>=3 Output: Differenz aufgrund abweichender Gewichte
Wackler	High-colli	signature	Condition: otherwise Output: Wackler rechnet Frachtrate für <reportTier>kg ab; wacklerRechnetFired only when FR<0
Wackler	Same-tier	signature	Condition: isBundle AND whole-kg AND weight gap > max(1kg,1%) Output: hätte gebündelt werden müssen
Wackler	Same-tier	signature	Condition: isBundle AND whole-kg AND weight gap <= max(1kg,1%) Output: Wackler rechnet Frachtrate für <reportTier>kg ab
Wackler	Same-tier	signature	Condition: volumetric (either weight non-integer) Output: Wackler rechnet Frachtrate für <floor-tier>kg ab; suppress TZ if fired
Wackler	Same-tier	signature	Condition: single reference Output: Wackler rechnet Frachtrate für <reportTier>kg ab; suppress TZ if fired
Wackler	No-weight	signature	Condition: multi-reference + FR significant Output: hätte gebündelt werden müssen

Appendix - exact shared constants and semantics

Item	Exact definition
hasErr	function hasErr(v,t){ return Math.abs(v) > t; }
Dachser tariff-zero marker	dalsTarifZero: blank => false; "-" => true; otherwise true when numeric value == 0 and raw text contains "0".
Dachser derived SNK	Non-integer SNK_DL only; rounded SNK_DIFF must be one of {5,9,11,14} and within 0.05 of rounded value.
K+N tier table	50..500 in 50kg steps; 600..2000 in 100kg steps; 2200..3000 in 200kg steps; 3500..7500 in 500kg steps; then 8500, 9500, 99999.
Wackler tier table	50..500 in 50kg steps; 600..2000 in 100kg steps; 2200..3000 in 200kg steps; 3500..10000 in 500kg steps; open >10000 bucket represented as 999999.
Wackler same-weight band	abs(VKG - VKG_DL) <= max(1kg, 1% of max weight).
Wackler bundle near band	A multi-reference cross-tier positive-FR row can bundle when weight gap <= 20% of the larger weight.
Wackler partial bundle	Cross-tier multi-reference terminal bundle when VKG_DL <= 20% of VKG.
Wackler high-colli synthetic weight	chargeable = max(colli*285, VKG, Brutto kg) for colli >= 7.
Wackler fuel threshold	TZ >= 2.0 is the additive energy-surcharge threshold; negative FR credits suppress the standalone fuel label.
Wackler Hebebuehne signature	SNK < 0 and abs(abs(SNK)-150) <= 2.0.
Wackler SNK noise floor	Unknown SNK only reaches generic fallback when abs(SNK) >= 5.0.
Current disabled rule	KONTIERUNG_ENABLED=false. Therefore KOST/SACH blank/X does not emit Kontierung? in the current build.

Verification context

Repository documentation states that the Anmerkung v5 integration has 215/215 regression tests passing and that the verification script checks source existence, Node syntax, required processors/helpers, the v5 facade, and the full repository test suite. The same documentation explicitly says this hardening result is not business-rule sign-off by itself.

Source note: the condition table is derived from the current executable source branches inspected in processDachser/daEvalSNK/daEvalEXP/processKN/processDHL/processWackler. Some output strings are generated from the PHRASES catalog; where the helper constant text was not repeated inline, the report uses the canonical phrase identified by the executable branch/tests.